

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 1: The Earth: Geometry, Coordinates & Distances

Syllabus Source: CAE Oxford (pp. 1-42) | Pass Target: $\geq 90\%$ AviationExam

1. EARTH DIMENSIONS & COORDINATE SYSTEM

The Earth is an oblate spheroid (flattened at the poles, bulging at the equator) modeled by the WGS-84 ellipsoid:

- Dimensions & Flattening: Equatorial Diameter = 12,756 km (6,888 NM); Polar Diameter = 12,714 km (6,865 NM). Compression / Flattening = (Equatorial - Polar) / Equatorial = 1 / 298.25.
- Latitude (Parallels): Angular distance North or South of Equator (0° to 90°). Colatitude = $90^\circ - \text{Latitude}$. Parallels are SMALL CIRCLES (except the Equator, which is a Great Circle).
- Longitude (Meridians): Angular distance East or West of Prime Meridian (Greenwich 000°) (0° to 180°). All meridians are SEMI-GREAT CIRCLES.
- The Nautical Mile Definition: 1 Nautical Mile (NM) is legally defined as EXACTLY 1,852 METERS (6,076.1 ft). Physically, 1 NM = the length of 1 MINUTE OF ARC ALONG A MERIDIAN of latitude!

2. THE DEPARTURE FORMULA (DISTANCE ALONG A PARALLEL)

Because meridians converge toward the poles, the linear east-west distance between two meridians decreases as latitude increases:

- The Master Departure Formula: Departure (NM) = Change of Longitude (in minutes of arc) $\times \cos(\text{Latitude})$
- Change of Longitude (dlong): dlong (minutes) = Departure (NM) / $\cos(\text{Latitude})$ = Departure $\times \sec(\text{Latitude})$

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating East-West Distance (Departure)

SCENARIO (Fundamental ATPL Navigation Drill): Two aircraft waypoints are located on the same parallel of latitude: - Waypoint A: $60^\circ 00' N$ $010^\circ 00' W$ - Waypoint B: $60^\circ 00' N$ $025^\circ 00' W$ QUESTION: What is the exact distance in Nautical Miles along the $60^\circ N$ parallel between Waypoint A and Waypoint B? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Change in Longitude (dlong): - Longitude A = $010^\circ W$ - Longitude B = $025^\circ W$ - Both are in the same hemisphere (West), so subtract: dlong = $25^\circ - 10^\circ = 15^\circ$. Step 2: Convert degrees of longitude into minutes of arc: - Rule: 1 degree = 60 minutes of arc. - dlong (minutes) = $15^\circ \times 60 = 900$ minutes of arc! Step 3: Recall the Master Departure Formula: - Formula: Departure (NM) = dlong (minutes) $\times \cos(\text{Latitude})$ Step 4: Plug in the numbers and calculate: - Latitude = $60^\circ N$. (Notice: $\cos(60^\circ) = 0.50$ exactly!). - Departure = $900 \text{ minutes} \times \cos(60^\circ) = 900 \times 0.50 = 450$ Nautical Miles! FINAL ANSWER: The distance along the $60^\circ N$ parallel is exactly 450 Nautical Miles (half of the 900 NM it would be at the equator!).

[!] EXAM TRAP (AVIATIONEXAM) - The $\cos(60^\circ) = 0.50$ Shortcut Trap

- At 60° latitude (North or South), $\cos(60^\circ)$ is EXACTLY 0.50! - This means at 60° latitude, 1° of longitude equals EXACTLY 30 NM (half of the 60 NM at the equator)! - Examiners LOVE 60° latitude questions because they can be solved instantly in your head without a calculator.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the distance in Nautical Miles along the parallel of 60°S between longitudes 020°E and 040°E ?**

[A] 1,200 NM [B] 600 NM ($20^\circ \times 60 \text{ min} = 1,200 \text{ min} \times \cos(60^\circ) = 600 \text{ NM}$) [C] 300 NM [D] 1,852 NM

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Change of longitude = $20^\circ = 1,200$ minutes. Departure = $1,200 \times \cos(60^\circ) = 1,200 \times 0.5 = 600 \text{ NM}$.

Q2: Which line on the Earth's surface is BOTH a parallel of latitude and a GREAT CIRCLE?

[A] Tropic of Cancer [B] THE EQUATOR (0° Latitude) [C] Greenwich Meridian [D] Arctic Circle

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The Equator is the only parallel of latitude whose plane passes directly through the center of the Earth, making it a Great Circle.

Q3: What is the international legal definition of ONE NAUTICAL MILE?

[A] 6,000 feet [B] Exactly 1,852 METERS (approx 6,076.1 ft) [C] 1,609 meters [D] 2,000 yards

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. By international agreement in 1929 (adopted by ICAO), 1 NM is defined as exactly 1,852 meters.